Financial Analysis of TATA MOTORS

Power BI Dashboard Project Report

Profit and loss trend, balance sheet structure, DAX measures and a supporting credit-risk dataset

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Tool: Microsoft Power BI Desktop (DAX)
Source file: TATA_Motors_Analysis.pbix

1. Project Overview

This project analyses the financial performance of Tata Motors using an interactive Power BI report. The report turns annual profit and loss and balance sheet data into KPI cards, trend charts, composition charts and a decomposition tree so that growth, profitability and balance sheet strength can be understood quickly.

Report structure (5 pages)

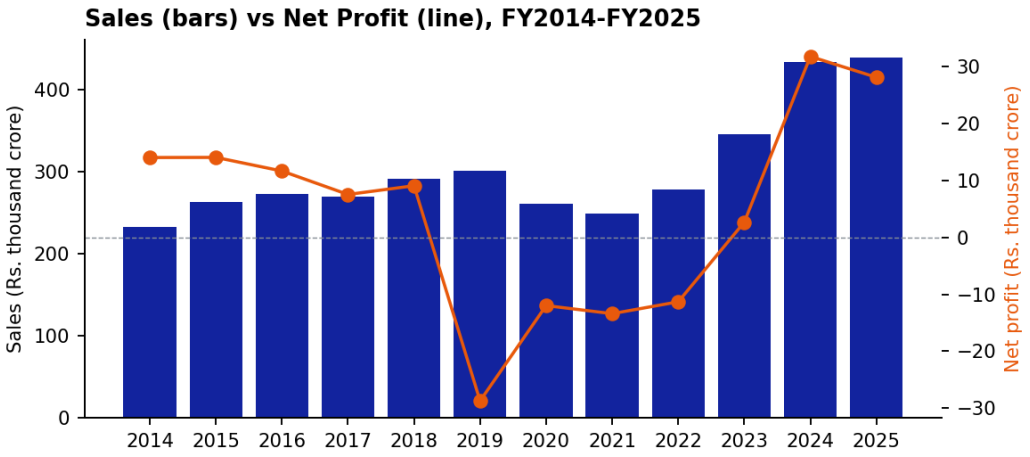
Page	Content
Page 1	Main dashboard: 3 combo charts (Reserves, Investments, Equity Capital by year), 3 KPI cards (Fixed Assets, Total Liabilities, Investments), gauge (Reserves), donut (Total Assets by year), pie (Fixed Assets by year), treemap, line chart (Total Assets) and a Year slicer
Page 2	Table of quarterly results (Sep 2024, Jun 2025, Sep 2025)
Page 3	Profit margin table by year plus notes on calculated column vs measure
Page 4	Clustered column chart of loan amount by loan grade (credit-risk dataset)
Page 5	Decomposition tree for drill-down analysis

Data model

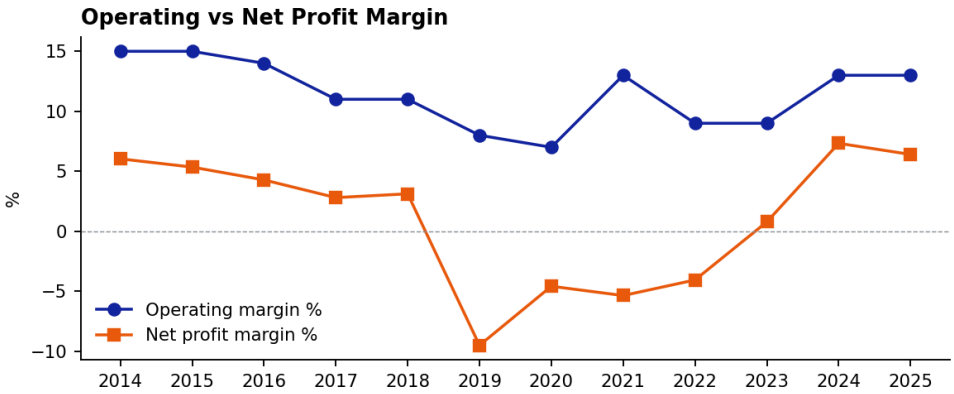
Table	Description	Rows
final	Annual P&L;, Mar 2014 to Mar 2025, plus TTM	13
Table 8	Annual balance sheet, Mar 2014 to Mar 2025	12
Table 2	Quarterly results (3 quarters)	3
creditriskdata	Loan applicant data (sample of 30 rows)	30

Note: figures appear to be in Rs. crore (Screener-style financial statements). The file does not state the unit, so please confirm it before publishing.

2. Profit and Loss Analysis



Sales grew from Rs. 232,834 crore in FY2014 to Rs. 439,695 crore in FY2025, a compound annual growth rate of about **5.9%**. Growth was flat to weak between FY2015 and FY2021, then accelerated: sales rose from Rs. 278,454 crore in FY2022 to Rs. 434,016 crore in FY2024.

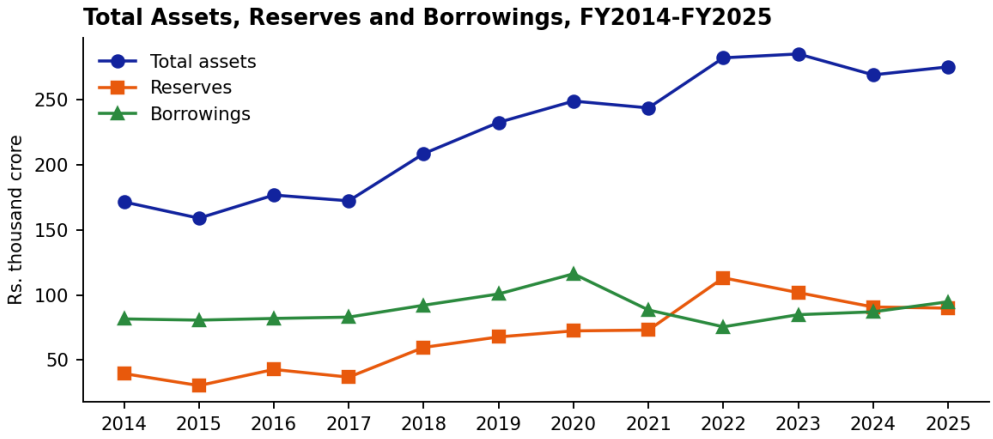


Net profit peaked at Rs. 31,807 crore in FY2024 and hit its low of Rs. -28,724 crore in FY2019. The company reported losses for four consecutive years (FY2019 to FY2022) before returning to profit in FY2023.

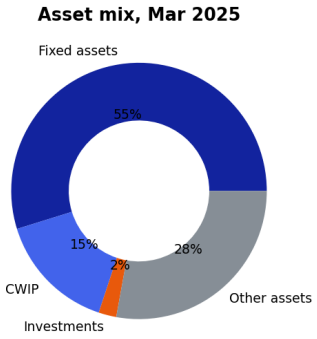
Annual P&L summary (Rs. crore)

Year	Sales	Op. Profit	OPM %	Interest	PBT	Net Profit	EPS (Rs)	Net margin %
2014	232,834	34,853	15	4,749	18,869	14,050	48.5	6.0
2015	263,159	39,239	15	4,861	21,703	14,073	48.4	5.3
2016	273,046	38,395	14	4,889	14,126	11,678	40.1	4.3
2017	269,693	29,589	11	4,238	9,315	7,557	25.8	2.8
2018	291,550	31,458	11	4,682	11,155	9,091	31.1	3.1
2019	301,938	24,664	8	5,759	-31,371	-28,724	-99.8	-9.5
2020	261,068	17,987	7	7,243	-10,580	-11,975	-39.1	-4.6
2021	249,795	32,287	13	8,097	-10,474	-13,395	-40.5	-5.4
2022	278,454	24,720	9	9,312	-7,003	-11,309	-34.5	-4.1
2023	345,967	31,816	9	10,225	3,394	2,690	7.3	0.8
2024	434,016	57,824	13	7,594	27,783	31,807	94.5	7.3
2025	439,695	55,216	13	5,083	38,651	28,149	75.6	6.4

3. Balance Sheet Analysis



Total assets increased from Rs. 171,604 crore (Mar 2014) to Rs. 275,459 crore (Mar 2025), growth of 61%. Reserves rose strongly to Rs. 113,222 crore in FY2022 and eased to Rs. 89,922 crore by FY2025. Borrowings fell from a peak of Rs. 116,328 crore in FY2020 to Rs. 75,561 crore in FY2022, then rose again to Rs. 94,801 crore.



Asset mix, Mar 2025

Fixed assets are the largest block at about 55% of total assets, with capital work in progress at 15%, indicating continued capacity investment. Investments account for only 2%.

Borrowings to net worth (equity + reserves) improved from 2.01x in FY2014 to 1.04x in FY2025 (low of 0.66x in FY2022).

Balance sheet summary (Rs. crore)

Year	Equity	Reserves	Borrowings	Other Liab.	Total Liab.	Fixed Assets	CWIP	Invest.	Total Assets
2014	971	39,561	81,629	49,443	171,604	74,907	26,822	5,093	171,604
2015	971	30,378	80,701	47,059	159,109	68,100	28,678	3,455	159,109
2016	970	42,762	81,987	51,165	176,884	72,200	35,996	10,714	176,884
2017	970	36,849	83,014	51,614	172,447	92,007	15,784	12,457	172,447
2018	1,145	59,726	92,147	55,704	208,722	96,105	16,614	17,899	208,722
2019	1,145	67,780	100,816	63,032	232,773	124,442	18,641	5,738	232,773
2020	1,145	72,431	116,328	59,245	249,149	134,551	19,497	6,285	249,149
2021	1,198	73,041	88,501	81,169	243,909	135,775	19,007	10,682	243,909
2022	1,221	113,222	75,561	92,417	282,422	133,288	22,046	13,140	282,422
2023	1,221	101,861	84,893	97,421	285,396	146,621	31,213	8,410	285,396
2024	1,247	90,788	87,082	90,195	269,312	148,814	34,356	6,258	269,312
2025	1,247	89,922	94,801	89,488	275,459	150,914	41,622	6,194	275,459

Total Liabilities equals Total Assets in every year, so the balance sheet reconciles.

4. Recent Quarterly Results

Quarter	Sales	Expenses	Op. Profit	OPM %	Other Inc.	PBT	Net Profit
Sep 2024	17,535	15,828	1,707	10	206	964	498
Jun 2025	17,324	15,248	2,076	12	332	1,674	1,397
Sep 2025	18,585	16,553	2,032	11	-1,865	-561	-867

Sep 2025 shows a net loss of Rs. 867 crore despite sales of Rs. 18,585 crore, driven by a negative other income of Rs. 1,865 crore. Operating profit itself stayed stable at Rs. 2,032 crore.

5. DAX: Calculated Column vs Measure

Profit margin was built in two ways to show the difference. A **calculated column** is computed row by row and stored in the model. A **measure** is evaluated at query time and responds to slicers and filters.

Calculated column (in the model)

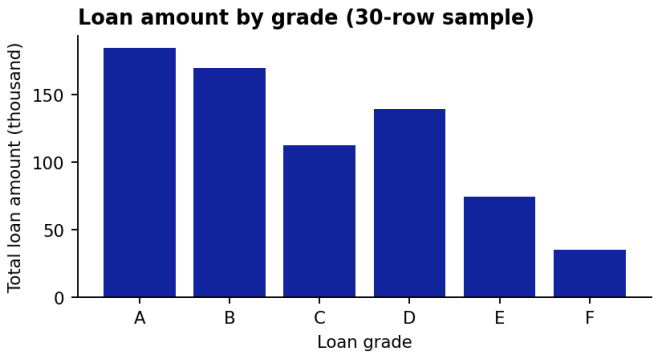
```
Rowmargincoloumn = DIVIDE('final'[Net Profit +], 'final'[Sales +])
```

Measures (documented on Page 3)

```
Total Profit = SUM(Income_Source[Profit_for_the_year_EUR_m])
Total Sales = SUM(Income_Source[Revenues_EUR_m])
RowProfitMargin = DIVIDE([Total Profit], [Total Sales])
```

The data model currently holds a measure named *Rowprofitmargin* on Table 2, alongside *profit* and *Sales*. The Page 3 notes refer to a table called Income_Source, which no longer exists in the model, so the notes should be updated to match the final table names.

6. Supporting Credit-Risk Dataset



Page 4 uses a loan dataset (age, income, home ownership, loan intent, grade, amount, interest rate, status). It is unrelated to Tata Motors, and the model contains only 30 rows, so the results below are illustrative.

Grade	Loans	Total amount	Avg. interest %	Status=1 share
A	9	184,900	8.1	44%
B	7	170,250	11.0	71%
C	5	112,500	13.6	100%
D	5	139,550	16.2	100%
E	3	74,375	17.6	67%
F	1	35,000	20.2	100%

Average interest rate rises steadily from about 8% (grade A) to 20% (grade F), as expected. Treat the status share cautiously because of the small sample.

7. Key Insights

- **Strong long-term growth:** sales grew at about 5.9% a year over FY2014-FY2025 and almost doubled between FY2021 and FY2025 (Rs. 249,795 crore to Rs. 439,695 crore).
- **Profitability recovered:** after four loss-making years (FY2019-FY2022) net profit returned to Rs. 31,807 crore in FY2024 and Rs. 28,149 crore in FY2025. Net margin is about 6.4% in FY2025.
- **Operating margin is stable:** OPM held at 13% in FY2024 and FY2025, compared with 7% to 9% in the weaker years.
- **Interest burden eased:** interest cost fell from Rs. 10,225 crore (FY2023) to Rs. 5,083 crore (FY2025).
- **Leverage improved but is rising again:** borrowings to net worth fell to 0.66x in FY2022 and stood at 1.04x in FY2025, with borrowings up from Rs. 84,893 crore in FY2023.
- **Heavy capital investment:** CWIP grew from Rs. 22,046 crore (FY2022) to Rs. 41,622 crore (FY2025).

8. Limitations and Next Steps

- Units are not stated in the file; confirm Rs. crore.
- The TTM row shows other income of Rs. 86,954 crore and net profit of Rs. 94,292 crore, which looks like a one-off item. It is excluded from the trend charts and should be checked against the source.

- Page 1 sums balance sheet items across years (for example the KPI cards and the donut), which has no business meaning. Use the year slicer or show a single year for these.
- Page 5 (decomposition tree) has no fields assigned in the saved file, so it appears empty.
- The credit-risk data is a separate, very small sample. Consider removing it or linking it to a clear purpose.
- Next steps: add ROE, ROCE and debt-to-equity measures, and add segment data (JLR vs India business) for deeper insight.

Source: data embedded in the Power BI file TATA_Motors_Analysis.pbix. This report is for academic purposes and is not investment advice.